

Business Requirements Document (BRD)

Apex Manufacturing Executive Dashboard

Project Name: Apex Manufacturing Executive Dashboard

Project Type: Business Intelligence Dashboard

Industry: Manufacturing

Tool: Microsoft Power BI

Prepared By: Hemanth Chelimala

Version: 1.0

Date: July 2026

1. Executive Summary

Apex Manufacturing Pvt. Ltd. is a manufacturing organization operating multiple production plants and managing various operational functions, including production, quality assurance, inventory management, maintenance, and procurement.

To improve operational visibility and support data-driven decision-making, the organization requires a centralized Business Intelligence dashboard that consolidates information from multiple departments into a single interactive reporting solution.

This project delivers a Power BI Executive Dashboard that enables management to monitor key business metrics, identify operational bottlenecks, and make informed strategic decisions through interactive visual analytics.

2. Business Background

Manufacturing organizations generate large volumes of operational data every day. These data sources are often maintained independently by different departments, making it difficult for management to obtain a unified view of business performance.

Apex Manufacturing requires an integrated dashboard capable of presenting production, quality, inventory, procurement, and maintenance information in a single platform to improve operational efficiency and executive reporting.

3. Business Problem

The organization currently experiences several reporting challenges:

- Departmental reports are maintained separately.
- Manual reporting consumes significant time.

- Management lacks real-time operational visibility.
- Performance comparisons across plants are difficult.
- Production and quality issues are identified too late.
- Inventory monitoring is inefficient.
- Maintenance costs are difficult to analyze.
- Procurement performance lacks centralized tracking.

These limitations reduce decision-making efficiency and increase operational risk.

4. Project Objectives

The primary objectives of this project are to:

- Develop a centralized Executive Dashboard.
 - Monitor manufacturing performance across all plants.
 - Track production efficiency and target achievement.
 - Analyze product quality and defect trends.
 - Monitor maintenance activities and machine performance.
 - Track inventory availability and stock levels.
 - Analyze procurement costs and supplier performance.
 - Enable interactive analysis using filters and slicers.
 - Support strategic decision-making through visual analytics.
-

5. Project Scope

The dashboard includes the following business domains:

Executive Dashboard

Provides a high-level overview of organizational performance through executive KPIs and summary visualizations.

Production Analysis

Analyzes production volume, target achievement, plant performance, and manufacturing efficiency.

Quality Analysis

Evaluates inspection results, defective units, rejection rates, and quality trends.

Maintenance Analysis

Monitors maintenance schedules, machine downtime, maintenance costs, and equipment utilization.

Inventory Analysis

Tracks stock availability, inventory movement, reorder status, and warehouse performance.

Procurement Analysis

Analyzes purchase orders, supplier performance, procurement costs, and purchasing trends.

6. Stakeholders

The dashboard is designed for the following stakeholders:

Stakeholder	Responsibility
Chief Executive Officer (CEO)	Strategic decision-making
Chief Operating Officer (COO)	Operational performance monitoring
Plant Managers	Production management
Production Managers	Manufacturing efficiency
Quality Assurance Team	Product quality monitoring
Maintenance Team	Equipment maintenance
Inventory Managers	Inventory optimization
Procurement Managers	Supplier and purchasing management
Business Analysts	Business performance analysis

7. Functional Requirements

The dashboard shall provide:

- Interactive slicers for filtering business data.
 - Dynamic KPI cards that respond to applied filters.
 - Cross-filtering between all visuals.
 - Trend analysis using monthly and yearly data.
 - Plant-wise performance comparison.
 - Product-wise production analysis.
 - Supplier performance analysis.
 - Inventory monitoring.
 - Maintenance cost analysis.
 - Executive-level summary reporting.
-

8. Non-Functional Requirements

The solution should satisfy the following quality requirements:

- Fast report performance.
 - Easy navigation across pages.
 - Consistent visual design.
 - Accurate KPI calculations.
 - Scalable data model.
 - Professional dashboard layout.
 - Easy future maintenance.
 - Reliable data refresh process.
-

9. Key Performance Indicators (KPIs)

The Executive Dashboard includes the following KPIs:

KPI	Description
Total Production	Total manufactured units
Production Target	Planned production quantity
Production Efficiency (%)	Production performance against target
Defective Units	Total defective products
Defect Rate (%)	Percentage of defective production
Inventory Stock	Current available inventory
Maintenance Cost	Total maintenance expenditure
Purchase Orders	Total procurement orders
Procurement Cost	Total purchasing expenditure
Active Suppliers	Number of suppliers contributing to procurement

10. Dashboard Features

The Power BI dashboard provides:

- Executive KPI Cards
- Interactive Slicers
- Dynamic Charts
- Monthly Trend Analysis
- Plant Performance Comparison
- Production Analysis

- Quality Monitoring
 - Maintenance Tracking
 - Inventory Analysis
 - Procurement Insights
 - Responsive Cross-Filtering
 - Business-Friendly Visualizations
-

11. Business Benefits

Implementation of this dashboard provides the following benefits:

- Centralized business reporting.
 - Improved operational transparency.
 - Faster executive decision-making.
 - Better production planning.
 - Early identification of quality issues.
 - Improved inventory control.
 - Reduced maintenance costs.
 - Better supplier evaluation.
 - Increased reporting efficiency.
 - Reduced manual reporting effort.
-

12. Deliverables

The project deliverables include:

- Microsoft Power BI Dashboard (.pbix)
 - Manufacturing Dataset
 - Business Requirements Document (BRD)
 - Data Dictionary
 - Data Profiling Report
 - Data Cleaning Report
 - Data Model Documentation
 - KPI Documentation
 - Dashboard User Guide
 - Final Project Report
 - GitHub Repository
-

13. Success Criteria

The project will be considered successful if:

- All manufacturing KPIs are available in a centralized dashboard.
- Users can interactively analyze data using slicers and filters.

- Dashboard calculations are accurate and validated.
 - Reporting time is significantly reduced.
 - Management can quickly identify operational trends and issues.
 - The dashboard supports data-driven business decisions.
-

14. Future Enhancements

Potential future improvements include:

- Integration with live ERP systems.
 - Automated scheduled data refresh.
 - Predictive analytics using machine learning.
 - Demand forecasting.
 - Production forecasting.
 - Row-Level Security (RLS).
 - Power BI Service deployment.
 - Mobile-optimized dashboard.
 - Real-time production monitoring.
-

15. Conclusion

The Apex Manufacturing Executive Dashboard transforms operational manufacturing data into meaningful business intelligence. By integrating production, quality, inventory, maintenance, and procurement data into a single reporting solution, the dashboard enables executives and operational managers to monitor key performance indicators, identify improvement opportunities, and make timely, data-driven decisions.

This project demonstrates the practical application of Microsoft Power BI for enterprise manufacturing analytics, combining effective data modeling, business-focused KPIs, and interactive visual reporting to improve organizational performance.